1. From two variables to three variables

So far, we have worked with probabilities involving one or two random variables. We know the sum rule, the product rule, and conditional probability.

The same rules apply when we have three variables. The only difference is that we have more possible outcomes to consider.

Suppose (C, T_1, T_2) are binary variables:

- $(C = 0)$: the person does not have cancer.
- $(C = 1)$: the person has cancer.
- $(T_1, T_2 \in 0, 1)$: the outcomes of two medical tests, where 1 means positive and 0 means negative.

There are $(2^3 = 8)$ possible combinations of these variables. For example,

$$(C, T_1, T_2) = (1, 0, 1)$$

means that the person has cancer, the first test is negative, and the second test is positive.

The **joint distribution** assigns a probability to each of these eight combinations:

$$P(C, T_1, T_2)$$

For example, $P(C = 1, T_1 = 0, T_2 = 1)$ is the probability that all three events occur together.

The probabilities of all eight combinations must sum to one. Therefore, an unrestricted joint distribution over three binary variables has $(8 - 1 = 7)$ independent probabilities. One probability is determined by the requirement that the total equals one.

More generally, an unrestricted joint distribution over (N) binary variables has

$$\boxed{2^N - 1}$$

independent probabilities.

This number grows very quickly as the number of variables increases. Graphical models help us represent joint distributions more efficiently by encoding conditional independence assumptions.

2. The product rule for three variables

Recall the product rule for two variables:

$$P(A, B) = P(A)P(B \mid A)$$

We can apply the same rule again to include a third variable.

First separate the T_2 from the joint distribution of the other two variables

$$P(C, T_1, T_2) = P(T_2 \mid C, T_1)P(C, T_1)$$

Then use the product rule again for $P(C, T_1)$

$$P(C, T_1) = P(T_1 \mid C)P(C) \$\$ Putting it together gives us \$\$ \begin{aligned} P(C, T_1, T_2) &= P(T_2 \mid C, T_1)P(C, T_1) \\ &= P(T_2 \mid C, T_1)P(T_1 \mid C)P(C \end{aligned}$$

This is the general product rule for the joint distribution of these three variables. This is ALWAYS true.

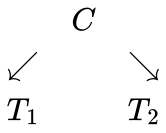
Notice that the last term is $(P(T_2 \mid T_1, C))$. In general, the outcome of the second test may depend on both the first test and the cancer status.

We cannot automatically remove (T_1) from this conditional probability. Doing so would require an additional assumption.

This is where the graphical model becomes important. The graphical model is the model of our specific domain.

3. The cancer graphical model

Consider the following graph:



The cancer status (C) is the common parent of the two test outcomes.

This graph denotes a particular factorisation of the joint probability distribution:

$$\boxed{P(C, T_1, T_2) = P(T_2|C)P(T_1|C)P(C)}$$

This factorisation is *not* true for a generic probability distribution, only for our specific domain (testing cancer). Compare it with the **domain independent** statement we proved in the previous section:

$$P(C, T_1, T_2) = P(T_2 \mid C, T_1)P(T_1 \mid C)P(C).$$

This shows us that

$$P(T_2 \mid T_1, C) = P(T_2 \mid C).$$

In other words, this graph encodes the assumtion that the two test results are conditionally independent given the cancer status:

$$P(T_2 \mid T_1, C) = P(T_2 \mid C).$$

If you *know* the patient cancer status, then the uncertainty in the test reduces to the test

accuracy, and test accuracy is assumed to be independent across different tests. Every time you test you get an independent error. This is what this graphical model encodes.

This factorisation is **not true** for every joint distribution. It encodes the conditional independence assumption represented by **this particular graph**.

What information do we need to perform Bayesian inference in this case?

To specify this model, we need:

- The prior probability, only one of $P(C = 0)$ or $P(C = 1)$
- The test probabilities $P(T_1 \mid C = 0)$ and $P(T_1 \mid C = 1)$.
- The test probabilities $P(T_2 \mid C = 0)$ and $P(T_2 \mid C = 1)$.

Since all variables are binary, each conditional distribution needs only one independent probability for each value of C. Together with $P(C)$, the model requires five independent parameters. If we assume T_1 and T_2 behave the same, i.e. they are the same type of test,

$$P(T_1 \mid C = 0) = P(T_2 \mid C = 0)$$

$$P(T_1 \mid C = 1) = P(T_2 \mid C = 1)$$

then we need only 3 independent variables.

Compare this with the seven independent parameters required for an unrestricted joint distribution over three binary variables.

The graph gives us a more compact representation because it encodes structure.

4. Independence versus conditional independence

The two tests are independent when the cancer status is known:

$$\boxed{P(T_1, T_2 \mid C) = P(T_1 \mid C)P(T_2 \mid C)}$$

For example, if we know that the person has cancer, the outcome of the first test does not change the probability of the second test, under our model.

However, if we do not know the cancer status, the two tests are generally dependent.

Why?

Each test result involves two sources of uncertainty:

1. Whether the person has cancer.
2. Whether the test gives a positive or negative result, given the person's true cancer status.

When the cancer status is unknown, learning that the first test is positive gives us information about whether the person has cancer. This changes our expectation about the second test.

Thus, in general,

$$P(T_2 \mid T_1) \neq P(T_2).$$

This is an example of **dependence caused by a common, unobserved cause**. You need the conditioning on the observation C to make them independent.

$$P(T_2 \mid T_1, C) = P(T_2 \mid C).$$

This is completely the opposite of the next example where the observation makes two independent variables dependent!

The tests have independent errors given the cancer status, but their outcomes are generally dependent when we average over the unknown cancer status.

5. A second example: Sunny, reward, and happiness

Now consider a different graph:

$$S \longrightarrow H \longleftarrow R$$

Here, S represents whether it is sunny, R represents whether someone receives a reward, and H represents whether the person is happy.

The graph encodes the factorization

$$\boxed{P(S, R, H) = P(S)P(R)P(H \mid S, R)}$$

The graph assumes that (S) and (R) are independent before observing happiness:

$$P(S, R) = P(S)P(R).$$

However, when we observe that the person is happy, S and R generally become dependent.

For example, if we learn that someone is happy, finding out that it is sunny may make a reward less necessary to explain their happiness. Conversely, if it is not sunny, we may consider a reward a more likely explanation.

This is called *explaining away*.

Compare the two graphs

In the cancer graph, C is a common cause of T_1 and T_2. The tests are generally dependent when C is unknown, but independent when C is known.

In the happiness graph, H is a common effect of S and R. The two causes are independent before observing H, but generally become dependent when H is observed.

These examples show why we must read the graph carefully. Different graph structures encode different independence assumptions.

6. How do we calculate probabilities we do not know directly?

The graph provides a compact representation of the joint distribution. Once we know the prior probabilities and the conditional probabilities in the model, we can calculate any probability involving the variables.

Some questions are straightforward. Others require us to eliminate variables that we do not observe.

The key operation is **marginalization**: sum over all possible values of a variable that we do not want to keep in the answer.

For example, suppose we want to calculate

$$P(T_1 = 1 \mid T_2 = 1).$$

We know the second test result, but we do not know whether the person has cancer. We therefore consider both possibilities, (C=0) and (C=1):

$$P(T_1 = 1 \mid T_2 = 1) = P(T_1 = 1, C = 0 \mid T_2 = 1) + P(T_1 = 1, C = 1 \mid T_2 = 1).$$

Note that this is the same as marginalziation, except that we have $T_2 = 1$ condition on **all** terms.
This equation is **always** true, and you can prove it with the three fundamental equations of normalization, sum and product rules.

This follows from the sum rule. The two events on the right are mutually exclusive and together cover all possibilities for the cancer status.

Now use conditional probability:

$$P(T_1 = 1, C = c \mid T_2 = 1) = \frac{P(T_1 = 1, C = c, T_2 = 1)}{P(T_2 = 1)}.$$

The graph tells us how to calculate the joint probabilities in the numerator:

$$P(T_1 = 1, C = c, T_2 = 1) = P(C = c)P(T_1 = 1 \mid C = c)P(T_2 = 1 \mid C = c).$$

We can therefore calculate both terms in the numerator using the probabilities specified by the model.

7. Calculating the denominator

We still need $P(T_2 = 1)$. We do not know the cancer status, so we sum over its possible values:

$$P(T_2 = 1) = P(T_2 = 1, C = 0) + P(T_2 = 1, C = 1).$$

Using the product rule and the graph's conditional independence assumptions,

$$P(T_2 = 1) = P(C = 0)P(T_2 = 1 \mid C = 0) + P(C = 1)P(T_2 = 1 \mid C = 1).$$

Substituting the numerator and denominator into our original expression gives

$$\boxed{\frac{\sum_{c \in 0,1} P(c)P(T_1 = 1 \mid c)P(T_2 = 1 \mid c)}{\sum_{c \in 0,1} P(c)P(T_2 = 1 \mid c)}}$$

The important point is not to memorize this formula. It is to understand the steps: use marginalization to account for the unknown cancer status, use the graph to factorize the joint probabilities, and then use conditional probability.

8. Another example: inferring cancer from a test

Suppose we observe a positive first test and want to calculate the probability that the person has cancer:

$$P(C = 1 \mid T_1 = 1).$$

By the definition of conditional probability,

$$P(C = 1 \mid T_1 = 1) = \frac{P(C = 1, T_1 = 1)}{P(T_1 = 1)}.$$

The numerator follows from the product rule:

$$P(C = 1, T_1 = 1) = P(C = 1)P(T_1 = 1 \mid C = 1).$$

For the denominator, marginalize over the cancer status:

$$P(T_1 = 1) = P(C = 0)P(T_1 = 1 \mid C = 0) + P(C = 1)P(T_1 = 1 \mid C = 1).$$

Therefore,

$$\boxed{\frac{P(C = 1)P(T_1 = 1 \mid C = 1)}{\sum_{c \in 0,1} P(c)P(T_1 = 1 \mid c)}}$$

This is Bayes' rule applied to the graphical model.

9. General strategy

Whenever we are given a graphical model and asked to calculate a probability, follow these steps.

1. **Read the graph.** Identify the variables and the joint probability distribution.
2. **Write the factorisation.** Express the joint distribution as a product of prior and conditional probabilities.
3. **Identify and marginalize the unknown variables.** If a variable is not part of the question and its value is unknown, sum over its possible values.
4. **Use the product rule.** Calculate the joint probabilities using the graph's factorisation.
5. **Use conditional probability or Bayes' rule.** Divide by the probability of the observed evidence when necessary.

The main lesson is that graphical models do not change the rules of probability. They use independence assumptions to represent joint distributions compactly and make complicated probability calculations easier.

10. Worked Example: Two medical tests and the probability of cancer

Suppose we have two identical medical tests, T_1 and T_2, that test for cancer, represented by C.

The variables are binary: $C = 1$ means the person has cancer, $C = 0$ means they do not, and $T_i = 1$ means test i is positive.

We know the following probabilities:

- $P(C = 1) = 0.01$: 1% of people have cancer.
- $P(T_i = 1 \mid C = 1) = 0.90$: each test is positive 90% of the time when a person has cancer.
- $P(T_i = 1 \mid C = 0) = 0.10$: each test is positive 10% of the time when a person does not have cancer.

We assume that the two tests are **conditionally independent given the cancer status**. This means that once we know whether a person has cancer, the result of one test does not affect the probability of the other test's result.

We will solve the same problem in two ways: first using natural frequencies, and then using probability formulas.

Method 1: Natural frequencies

Imagine testing 10,000 people.

Since 1% of people have cancer, 100 people have cancer and 9,900 do not.

Test outcome	Cancer	No cancer	Total
First test positive	90	990	1,080

Test outcome	Cancer	No cancer	Total
Both tests positive	81	99	180

Let us see how these numbers arise.

Among the 100 people with cancer, 90 test positive on the first test. Of these 90 people, 81 also test positive on the second test.

Among the 9,900 people without cancer, 990 test positive on the first test. Of these 990 people, 99 also test positive on the second test.

Question 1: What is the probability that a person has cancer given that their first test is positive?

There are 1,080 people with a positive first test, and 90 of them have cancer. Therefore,

$$P(C = 1 \mid T_1 = 1) = \frac{90}{1080} \approx 0.0833$$

The probability is approximately **8.33%**.

Notice that a positive test does not mean that cancer is likely. Because cancer is rare, there are many more false positives than true positives.

Question 2: What is the probability that a person has cancer given that both tests are positive?

There are 180 people with two positive tests, and 81 of them have cancer. Therefore,

$$P(C = 1 \mid T_1 = 1, T_2 = 1) = \frac{81}{180} = 0.45$$

The probability is **45%**.

The second positive test increases the probability of cancer from approximately 8.33% to 45%.

Method 2: Probability formulas

We will now calculate probabilities using the graphical model and the rules of probability.

The graph has cancer status as the common cause of the two test outcomes:

$$C \longrightarrow T_1 \qquad C \longrightarrow T_2$$

The graph encodes the following factorisation:

$$P(C)P(T_1 \mid C)P(T_2 \mid C)$$

This means that the joint probability can be calculated by multiplying the prior probability of cancer by the conditional probabilities of the two test outcomes.

Step 1: Calculate the probability of cancer given one positive test

We want to calculate $P(C = 1 \mid T_1 = 1)$.

We have to write everything in terms of what we know:

- $P(C = 1) = 0.01$: 1% of people have cancer.
- $P(T_i = 1 \mid C = 1) = 0.90$: each test is positive 90% of the time when a person has cancer.
- $P(T_i = 1 \mid C = 0) = 0.10$: each test is positive 10% of the time when a person does not have cancer.
- Our problem specific factorization: $P(C, T_1, T_2) = P(C)P(T_1 \mid C)P(T_2 \mid C)$

Using conditional probability,

$$P(C = 1 \mid T_1 = 1) = \frac{P(C = 1, T_1 = 1)}{P(T_1 = 1)}$$

This is always true, not only for our graph.

Using the product rule, the numerator is

$$P(C = 1, T_1 = 1) = P(C = 1)P(T_1 = 1 \mid C = 1)$$

Substituting the numbers,

$$P(C = 1, T_1 = 1) = 0.01 \times 0.9 = 0.009$$

We still need the denominator, $P(T_1 = 1)$. But all that we know are about how T_1 relate to C. This is the most creative step. The strategy is to introduce C and then get rid of it by marginalization! We do not know the cancer status, so we sum over both possible values of C.

$$P(T_1 = 1) = P(C = 0, T_1 = 1) + P(C = 1, T_1 = 1)$$

Now we can use the product rule to get:

$$P(T_1 = 1) = P(C = 0)P(T_1 = 1 \mid C = 0) \ + P(C = 1)P(T_1 = 1 \mid C = 1)$$

Substituting the numbers,

$$P(T_1 = 1) = 0.99 \times 0.1 + 0.01 \times 0.9 \qquad = 0.099 + 0.009 \ = 0.108$$

Therefore,

$$P(C = 1 \mid T_1 = 1) = \frac{0.009}{0.108} \qquad \approx 0.0833$$

The probability is approximately **8.33%**, matching the natural-frequency calculation.

Take a moment to note that *we needed nothing more than the basic probability rules we learned in Session One!* You do not need to memorise a whole slew of formulas, just learn a few key principles and get skilled in manipulating them.

Step 2: Calculate the probability of a positive first test given a positive second test

Now suppose we know that the second test is positive. We want to calculate

$$P(T_1 = 1 \mid T_2 = 1)$$

Using Bayes rule we have

$$P(T_1 = 1 \mid T_2 = 1) = \frac{P(T_1 = 1, T_2 = 1)}{P(T_2 = 1)}$$

But how can we compute the terms in the numerator and denominator? Remember our strategy from the previous section. The cancer status is unknown, so we must consider both possibilities: $C = 0$ and $C = 1$.

Using the sum rule,

$$P(T_1 = 1, T_2 = 1) = P(C = 0, T_1 = 1, T_2 = 1) + P(C = 1, T_1 = 1, T_2 = 1)$$

and

$$P(T_2 = 1) = P(C = 0, T_2 = 1) + P(C = 1, T_2 = 1)$$

I leave the rest to you to solve. Try to use what we know to compute these terms, then compare them against the results you found numerically.

Step 3: What does this tell us about independence?

Three probabilities, three different meanings

Consider the following three probabilities:

A. $P(T_2 = 1 \mid C = 1) = 0.9$

The probability that the second test is positive for someone who has cancer.

B. $P(T_2 = 1) = 0.108$

The probability that the second test is positive for a randomly selected person from the population, without knowing whether they have cancer.

C. $P(T_2 = 1 \mid T_1 = 1) \approx 0.1667$

The probability that the second test is positive for a randomly selected person whose first test was positive, without knowing whether they have cancer.

Can you explain the differences between these three probabilities?

What information do we have in each case, and how does that information affect the probability of a positive second test result?

Summary

We solved the same problem using two methods. Natural frequencies gave us an intuitive way to understand the results, while probability formulas showed how to derive them from the graphical model.

The important steps in the formula-based method were to use the graph to factorise the joint probability, sum over the unknown cancer status, and apply conditional probability.

You do not need to memorise a separate formula for every question. Focus on understanding which probability rules are being used and why.

Practice question: Sunny, raise, and happiness

Consider three binary variables:

- $S = 1$: the weather is sunny; $S = 0$: it is not sunny.
- $R = 1$: a person receives a raise; $R = 0$: they do not receive a raise.
- $H = 1$: the person is happy; $H = 0$: they are not happy.

We model their relationships using the following graphical model:

$$S \longrightarrow H \longleftarrow R$$

The joint distribution factorises as

$$P(S, R, H) = P(S)P(R)P(H \mid S, R).$$

Suppose we know the following probabilities.

***Question** This model assumes that S and R are independent, why?*

Prior probabilities

$$P(S = 1) = 0.5, \qquad P(R = 1) = 0.3$$

Assume that sunny weather and receiving a raise are independent before observing happiness. Therefore,

$$P(S, R) = P(S)P(R).$$

Conditional probabilities of happiness

The probability of being happy depends on both the weather and whether the person receives a raise.

Weather	Raise	Probability of happiness
Sunny ($S = 1$)	Raise ($R = 1$)	$P(H = 1 \mid S = 1, R = 1) = 0.95$
Sunny ($S = 1$)	No raise ($R = 0$)	$P(H = 1 \mid S = 1, R = 0) = 0.80$
Not sunny ($S = 0$)	Raise ($R = 1$)	$P(H = 1 \mid S = 0, R = 1) = 0.85$
Not sunny ($S = 0$)	No raise ($R = 0$)	$P(H = 1 \mid S = 0, R = 0) = 0.10$

***Question** Why we do not have to specify $P(H = 0 \mid S = 1, R = 1)$ etc...?*
***Question** Why the probabilities for happiness do NOT sum up to one?*

Part A: From probabilities to frequencies

Imagine observing 1,000 randomly selected days.

1. How many days would you expect to be sunny?
2. How many days would you expect the person to receive a raise?
3. How many days would you expect it to be both sunny and a day on which the person receives a raise?
4. How many days would you expect the person to be happy, the weather to be sunny, and the person to receive a raise?

Part B: Calculate the probability of happiness

Calculate the probability that the person is happy:

$$P(H = 1).$$

The person could be happy in any of the four possible combinations of weather and raise status. Use the sum rule to account for all four cases.

Start by writing

$$P(H = 1) = \sum_{s \in 0,1} \sum_{r \in 0,1} P(H = 1, S = s, R = r).$$

Then use the product rule and the graphical model to express each joint probability in terms of the given probabilities.

Part C: Explaining away

Suppose you learn that the person is happy, so $H = 1$.

First, calculate the probability that the person received a raise given that they are happy:

$$P(R = 1 \mid H = 1).$$

Then calculate the probability that the person received a raise given that they are happy and that it was sunny:

$$P(R = 1 \mid H = 1, S = 1).$$

Compare your answers. Does knowing that it was sunny change how likely you think it is that the person received a raise, given that they are happy?

Explain your result using the graphical model and the idea of *explaining away*.

Show your working using the sum rule, product rule, and conditional probability. Do not assume that S and R remain independent after observing H.